In one of our last interactions, Ripper said, ‘Sorry sir, I can’t be of much help to you, but I am sure you will soon walk free.’

HOME NOT SWEET

‘It may remain a dream; it may not take off at all.’

Problems arising out of internal politics, and the continuing stepmother treatment to liquid propulsion system, awaited me in India. I anticipated them, but did not allow them to weigh me down. In fact, before I packed my bags for India in March 1978, I had set my agenda to identifying a place and get infrastructure for a liquid propulsion centre; get project approval for Vikas, the liquid engine; get into the Polar Satellite Launch Vehicle (PSLV) project, which Dhawan had outlined by making Vikas a part of it.

Even as Kalam’s SLV project was on stream, Dhawan had initiated planning for a PSLV and a GSLV (Geosynchronous Satellite Launch Vehicle). While PSLV and GSLV were still in the ideation stage, clearance had been given for a project parallel to SLV, the Augmented Satellite Launch Vehicle (ASLV). M S R Dev, the ASLV project director, was an efficient scientist, a PhD from the US. But the vehicle was aerodynamically jinxed so much that every time it was test-launched it plunged into the sea. In ISRO, ASLV soon came to be known as ‘Always Sea-Loving Vehicle’.

That was a cruel joke, given the effort Dev and his team had put into the ASLV project. Today, with six-dimensional trajectory analysis software, we are able to accurately predict flight load, trajectory, overflying countries, areas of debris fall and much more.

With these modern tools, we now foresee problems through simulation, and prevent them before the launch. Back in the late 1970s, we were not good at 6D trajectory analysis, and it was all trial and error. ASLV may not be a shining showpiece in the ISRO brochures today, but our success with